UEE413 : ELECTRIC MACHINERY

L	T	P	Cr
3	1	2	4.5

Course Objectives: The course introduces the concept of D.C. Machines, Transformers, Synchronous & Asynchronous AC machines, and fractional **H.P.** motors, as well as their construction and performance parameters.

Syllabus

Direct Current Machines: Basic concept and classification of dc machines as per type of excitation, circuit models and related equations of separately and self-excited dc generators and motors, armature reaction, characteristics of dc generators, characteristics of dc motors , speed control of dc motor, DC Motor starters, losses and efficiency in DC machines .

Three Phase Transformers: Working principle **and construction** of three phase transformers, construction, basic phasor groups and connections of three phase transformers, parallel operation of three-phase transformers.

Asynchronous Machines (Induction Motors): Principle of operation and construction of squirrel cage and slip-ring induction motors, calculation of slip, rotor frequency, rotor emf, output power losses and efficiency, induction motor phasor diagram and equivalent circuit, torque-slip and power-slip characteristics, determination of equivalent circuit parameters from no-load test and blocked-rotor test, starting methods of induction motor, methods of speed control.

Synchronous Machines: Operating principle construction; **and** phasor diagrams of cylindrical and salient pole synchronous generators/alternator, Excitation systems, Open circuit and short circuit test of synchronous machine, voltage regulation of an alternator using synchronous impedance method, active and reactive power control methods for synchronous generators and motors, power-angle characteristics , Parallel Operation of alternators and Synchronisation of generator with infinite bus, working principle and application of synchronous motor.

Fractional Horse Power Motors: Basic concept of single-phase induction motor, starting methods, comparison between single and poly-phase induction motors, working principle, **construction** and application of universal motor, single-phase reluctance motor, sub-synchronous motor, hysteresis motor.

Laboratory Work

Minor project: Modeling and simulation of electric machines using software tools.

- Obtain the magnetization curve of a D.C. machine.
- Perform load tests on separate and self-excited DC generators.
- Perform speed control of a D.C. shunt motor.
- Perform parallel operation of two single-phase transformers.
- Learn three-phase transformer connections.
- Learn the connection of starters for a three-phase Induction motor.
- Perform no-load and blocked-rotor tests to determine equivalent circuit parameters of a three-phase induction motor.
- Obtain the performance characteristics of a three-phase induction motor.
- To determine alternator voltage regulation using the synchronous impedance method.
- Perform synchronization of an alternator with the infinite bus.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. **Demonstrate three-phase transformer connections and justify load sharing among parallel-connected three phase transformers**
2. Analyse the performance of DC **generators** and DC **motors**
3. Analyse the performance of Three-phase induction motor
4. **Critically analyze voltage regulation and load sharing of synchronous generators**
5. **Select applications of fractional kilowatt motors**

Text Books:

1. D.P. Kothari and I.J. Nagrath, Electric Machines, 4e, Tata McGraw Hill Education Private Limited, New Delhi.
2. P.S. Bimbhra, Electrical Machinery, 7ed., Khanna Publishers, New Delhi.
3. P.S. Bimbhra, Generalized Theory of Electrical Machines, 5e, Khanna Publishers, New Delhi.

Reference Books:

1. Toro, Vincert, Electromechanical Devices for Energy Conversion, Prentice Hall of India.
2. Fitzgerald, A.E., Kingsley, C. Jr., and Umans, Stephen, Electric Machinery, 6e, McGraw Hill, USA.

Evaluation Scheme

Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	**70**
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	**30**